Redox mechanisms: Exercises

> Reference: Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

All numerical cases are hypothetical, with the assumptions stated in each question. Give units and identify where a model would fail.

E01 Reaction and neutrality

For $O_O^{\times} \rightarrow \frac{1}{2}O_2 + V_O^{\bullet\bullet} + 2e'$, verify oxygen-atom count and effective charge. If electrons and doubly positive vacancies are the only charged species and local neutrality holds, find electron concentration when the vacancy concentration is $2 \times 10^{18}\,\mathrm{cm^{-3}}$. Name two reasons this estimate can fail.

E02 Drift and diffusion

In a dilute isothermal one-dimensional model, $J = -D\,dc/dx + (Dze/k_BT)cE$. Take $D = 10^{-18}\,\mathrm{m^2/s}$, $z = 2$, $E = 10^{8}\,\mathrm{V/m}$, $c = 10^{24}\,\mathrm{m^{-3}}$, $dc/dx = 10^{30}\,\mathrm{m^{-4}}$, and $k_BT = 4.14 \times 10^{-21}$ J. Use $e = 1.60 \times 10^{-19}$ C. Compute diffusion, drift and net number fluxes, including signs and units.

E03 Potential and migration barrier

A 12 nm film has a uniform 0.6 V drop. Calculate field magnitude, signed potential difference across a 0.2 nm hop along the electric field, and signed electrostatic endpoint energy difference for $z = +2$. Can that number alone determine the activation-barrier reduction? Explain.

E04 Gap sensitivity

In the hypothetical fixed-barrier model $I(g) = I_0 e^{-2\kappa g}$, $\kappa = 5\,\mathrm{nm^{-1}}$. Find $I(1.3\,\mathrm{nm})/I(1.0\,\mathrm{nm})$. Give two effects ignored by the model.

E05 Thermal transient

A lumped device starts at ambient with constant $P = 0.1$ mW, $R_{\mathrm{th}} = 10^{6}$ K/W and $C_{\mathrm{th}} = 10^{-13}$ J/K. Calculate the thermal time constant, steady-state temperature rise and temperature rise at $t = 100$ ns. Why is energy alone insufficient to find the peak temperature?

Lin Cing https://github.com/xinhi6716/resistive-memory-self-study

Redox mechanisms: Exercises (continued)

E06 Activated timescale

Assume one thermally activated process at constant 300 K obeys $\tau = 10^{-12}\exp(E_a/k_BT)$ s, with $E_a = 1.0$ eV and $k_BT = 0.02585$ eV. Estimate τ in seconds and hours. What additional information is needed before claiming a retention time for a real memory?

E07 Mechanisms and boundaries

An Ag-based cell exhibits switching. Write an oxidation and a reduction half-reaction and identify where electrons and mobile ions may enter your explanation. Does this information determine the direction from which a filament grows? Contrast a VCM alternative and name two measurements that can help distinguish models.

E08 Hysteresis versus retention

A triangular voltage sweep produces an I–V loop passing through the origin. Design a controlled experiment that tests nonvolatile retention and explain why the crossing alone is insufficient. Name two possible causes of a nonzero current at zero applied terminal voltage.

E09 Boundary flux and threshold

An idealized $L = 10$ nm slab has a spatially averaged defect concentration $N(0) = 2 \times 10^{18}\,\mathrm{cm}^{-3}$. Assume constant injection and removal fluxes $s = 8 \times 10^{12}$ and $r = 2 \times 10^{12}\,\mathrm{cm}^{-2}\mathrm{s}^{-1}$, with no volume reactions. When does the average reach a model threshold $N_c = 5 \times 10^{18}\,\mathrm{cm}^{-3}$? Does this establish a local metallic filament?

E10 Compare same-polarity scan energy

Two positive voltage ramps start at zero with $dV/dt = 1$ V/s. SET has $I(V) = 1\,\mu\mathrm{S} \times V$ until 1 V; RESET has $I(V) = 4\,\mu\mathrm{S} \times V$ until 0.5 V. Integrate the energy delivered in each ramp, and explain whether their peak temperatures follow from the energy comparison alone.

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Redox mechanisms: Exercises (continued)

E11 Local concentration and threshold interval

A one-dimensional region of thickness $L = 10$ nm has fixed $c(0) = 8 \times 10^{18}$ and $c(L) = 2 \times 10^{18}\,\mathrm{cm}^{-3}$. Assume constant temperature, $D = 10^{-14}\,\mathrm{cm}^2/\mathrm{s}$, no drift or volume sources, and steady state. Use $J = -D dc/dx$ to find $c(x)$ and the flux magnitude and direction. If $N_c = 5 \times 10^{18}\,\mathrm{cm}^{-3}$, find the interval and length where $c(x) \geq N_c$. Does this establish a connected filament?

E12 Acceptor compensation

Suppose $[V_O^{\bullet\bullet}] = 3 \times 10^{18}\,\mathrm{cm}^{-3}$ and the concentration of singly negative ionized acceptors is $2 \times 10^{18}\,\mathrm{cm}^{-3}$. Assume local neutrality and no other charged species. Find the electron concentration, compare the result obtained if acceptors were omitted, and state where local neutrality may fail.

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